

SCOTT (SEONGWON) LEE

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SUMMARY

- **Robotics and AI** researcher focused on multi-robot planning, foundation models for robotics, neuro-symbolic reasoning, and trust-worthy autonomous systems.
- Developed **hypergraph-based multi-robot task and motion planning (MR-TAMP)** algorithms for complex robotic systems, with work published in *IEEE Transactions on Robotics* and validated on physical multi-robot platforms. (NSF-funded project)
- Built **agentic LLM-driven robotic automation systems at X, The Moonshot Factory (formerly Google X)**, translating natural-language experimental protocols into verified robot plans and executable code for autonomous biological laboratory workflows.

EDUCATION

University of Illinois Urbana-Champaign

Aug. 2021 – Dec. 2026 (Expected)

Ph.D. Candidate in Mechanical Science and Engineering

Advisor: Prof. Nancy M. Amato

University of Illinois Urbana-Champaign

Aug. 2021 – Dec. 2024

M.S. in Mechanical Engineering

Advisor: Prof. Nancy M. Amato

Yonsei University, Seoul, South Korea

Mar. 2016 – Feb. 2021

B.S. in Mechanical Engineering

Advisor: Prof. Jongeun Choi

- National Science and Technology Scholarship (full tuition); graduated *cum laude*.

University of California, Berkeley

Jan. 2020 – May 2020

Exchange Student Program

RESEARCH EXPERIENCE

X, The Moonshot Factory (formerly Google X)

May 2025 – Dec. 2025

Ph.D. Resident

- Built an **LLM-agent orchestration system** to translate natural-language wet-lab protocols into robot-executable workflows and code for Hamilton liquid handlers and Universal Robots manipulators.
- Developed a **generation–verification–execution pipeline** that validates LLM-generated robot actions and code using simulation and formal planning before execution on physical robotic systems.
- Integrated agentic reasoning with **robotic task and motion planning** to autonomously plan, verify, and execute multi-step biological laboratory workflows.

Parasol Lab, University of Illinois Urbana-Champaign

May 2022 – Present

Ph.D. Student (Advisor: Prof. Nancy M. Amato)

• Scalable Multi-Robot Task and Motion Planning

- Designed **hypergraph-based MR-TAMP algorithms** for scalable planning across increasing robot, object, and task counts.
- Developed hierarchical planning methods that combine task planning, motion feasibility, and constraint reasoning for complex multi-robot systems.
- Validated the planning methods on physical multi-robot automation platforms and in an **NSF-funded biological laboratory automation** project.

• LLM/VLM-Driven Robotic Automation

- Developing **vision-language-guided multi-robot task and motion planning** methods for online reasoning and replanning.
- Integrating foundation model reasoning with structured symbolic and geometric planning for reliable execution in physical environments.

• Open-Source Robotics Software

- Contributed to the open-source C++ **Parasol Planning Library (PPL)** for robot task and motion planning ([GitHub](#)).

Machine Learning and Control Systems Lab, Yonsei University

May 2018 – May 2020

Undergraduate Researcher (Advisor: Prof. Jongeun Choi)

- Developed **CNN-DDPG autonomous-driving methods** for unexpected collision avoidance and vehicle control.
- Developed robust control and state-estimation methods for **UAVs and quadrotors**, combining dynamic inversion, recursive least squares, and high-gain observers.

Deep Machine Lab (Startup, acquired by Brand Engagement Network) (Webpage)

Feb. 2021 – Jun. 2021

Machine Learning Researcher (Supervisor: Prof. Hanseok Ko)

- Developed a **human-interactive virtual avatar** using gesture-generation networks for natural human–avatar interaction.

PUBLICATIONS

Reverse chronological order

÷ Non-peer-reviewed • Peer-reviewed

- **Seongwon Lee**, James Motes, Isaac Ngui, Marco Morales, Nancy M. Amato, "PDDL DaSH: Hypergraph-based Multi-robot Task Planning" **In preparation**
- **Seongwon Lee**, James Motes, Isaac Ngui, Marco Morales, Nancy M. Amato, "Lazy DaSH: Lazy Approach for Hypergraph based Multi-robot Task and Motion Planning" **IEEE Transactions on Robotics**
- ÷ **Seongwon Lee**, James Motes, Isaac Ngui, Marco Morales, Nancy M. Amato, "Lazy DaSH: Lazy Approach for Hypergraph based Multi-robot Task and Motion Planning" **ICRA@40 Extended Abstract**
- ÷ Isaac Ngui, **Seongwon Lee**, James Motes, Marco Morales, Nancy M. Amato, "A hierarchical Approach to Workstation-based Task Allocation and Motion Planning" **IROS 2023 Workshop Paper**
- **Seongwon Lee**, Joohwan Seo, Connor J. Boss, Joonho Lee, Jongeun Choi, "Output Feedback Control Design for Quadrotors Using Recursive Least Square Dynamic Inversion" **Elsevier Mechatronics**
- Myunhoe Kim, **Seongwon Lee**, Jaehyun Lim, Jongeun Choi, Seong Gu Kang, "Unexpected Collision Avoidance Driving Strategy Using Deep Reinforcement Learning" **IEEE Access**
- Joohwan Seo, **Seongwon Lee**, Joonho Lee, and Jongeun Choi, "Nonaffine helicopter control design and implementation based on a robust explicit nonlinear model predictive control" **IEEE Transactions on Control System Technology**

TEACHING & MENTORSHIP

- **Research Program Mentor** – Computer Science, UIUC *Fall 2023 – Present*
- **Teaching Assistant, ME 310** – Fluid Mechanics *Fall 2022, Spring 2024, Fall 2024*
- **Teaching Assistant, TAM 541** – Mathematical Methods *Fall 2023*
- **Engineering Open House (EOH) (Webpage)** *Apr. 2024*
 - Presented a factory-automation demonstration to middle and high school students

AWARDS & COMPETITIONS

ICRA 2019 DJI RoboMaster AI Challenge (Project Description)

- Achieved **3rd place among 32 selected teams worldwide**.
- Implemented a **Multi-Agent DDPG (MADDPG)** algorithm for collaborative maneuvering.

BMW Research Competition

- Achieved **1st place** in a competition hosted by BMW Korea.
- Developed a **CNN-DDPG autonomous-driving controller** for unexpected collision avoidance.

TECHNICAL SKILLS

Programming	Python, C++
Agentic AI	LangGraph, Google Agent Development Kit (ADK)
Robotics & Simulation	ROS/ROS2, Gazebo, MuJoCo, Unreal Engine, Simulink
Hardware	Universal Robots manipulators, Intel RealSense, NVIDIA Jetson, Raspberry Pi, Arduino
Modeling	Autodesk Fusion 360, Creo

LEADERSHIP EXPERIENCE

- **Military Service, Republic of Korea** *Mar. 2014 – Dec. 2015*
- **Student President, School of Mechanical Engineering, Yonsei University** *Mar. 2016 – Dec. 2016*